BRZOZOWSKI AND CALCULUS DERIVATIVES

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Consider regular expressions over the alphabet $\Sigma = \{x, y\}$:

$$e ::= 0 \mid 1 \mid x \mid y \mid e_1 + e_2 \mid e_1 e_2 \mid e^*$$

The Brzozowski derivative of a regular expression e with respect to a symbol x is defined as:

$$\begin{aligned}
\delta_x(0) &= 0 \\
\delta_x(1) &= 0 \\
\delta_x(x) &= 1 \\
\delta_x(y) &= 0 \text{ if } x \neq y \\
\delta_x(e_1 + e_2) &= \delta_x(e_1) + \delta_x(e_2) \\
\delta_x(e_1 e_2) &= \delta_x(e_1) e_2 + \epsilon(e_1)\delta_x(e_2) \\
\delta_x(e^*) &= \delta_x(e) e^*
\end{aligned}
\qquad
\begin{aligned}
\epsilon(0) &= 0 \\
\epsilon(1) &= 0 \\
\epsilon(x) &= 0 \\
\epsilon(y) &= 0 \\
\epsilon(e_1 + e_2) &= \epsilon(e_1) + \epsilon(e_2) \\
\epsilon(e_1 e_2) &= \epsilon(e_1)\epsilon(e_2) \\
\epsilon(e^*) &= \epsilon(e)^*
\end{aligned}$$

When we expand e^* we get:

$$e^* = 1 + e + e^2 + e^3 + \cdots$$

As a power series in calculus, we have:

$$\frac{1}{1-e} = 1 + e + e^2 + e^3 + \cdots$$

Therefore, e^* is analogous to $\frac{1}{1-e}$. Let us therefore define e^* as $\frac{1}{1-e}$ on the calculus side. Note that the derivative of $\frac{1}{1-e}$ is $\frac{e'}{(1-e)^2}$. We therefore get the following calculus rules:

$$\begin{aligned}
\delta_x(0) &= 0 \\
\delta_x(1) &= 0 \\
\delta_x(x) &= 1 \\
\delta_x(y) &= 0 \text{ if } x \neq y \\
\delta_x(e_1 + e_2) &= \delta_x(e_1) + \delta_x(e_2) \\
\delta_x(e_1 e_2) &= \delta_x(e_1) e_2 + e_1 \delta_x(e_2) \\
\delta_x(e^*) &= \delta_x(e)(e^*)^2
\end{aligned}$$

Note the difference in the rules:

Brozowski:

$$\begin{aligned}
\delta_x(e_1 e_2) &= \delta_x(e_1) e_2 + \epsilon(e_1)\delta_x(e_2) \\
\delta_x(e^*) &= \delta_x(e) e^*
\end{aligned}$$

Calculus:

$$\begin{aligned}
\delta_x(e_1 e_2) &= \delta_x(e_1) e_2 + e_1 \delta_x(e_2) \\
\delta_x(e^*) &= \delta_x(e)(e^*)^2
\end{aligned}$$

Therefore, although we call both derivatives, they are quite different.